Exercises for Geometric Representation Theory

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1 Introduction

Exercise. Prove the following statements:

1. All choices of Hermitian inner product on $\mathbb{C}^2$ are conjugate. So $\mathrm{SU}_2 \to \mathrm{SL}_2$ is defined up to conjugation.
2. $\mathrm{SU}_2 = S^3$
3. $\mathrm{SL}_2(\mathbb{C}) \cong T\mathrm{SU}_2 = TS^3$ diffeomorphism. In particular SU_2 is a maximal compact subgroup of $\mathrm{SL}_2(\mathbb{C})$, and all maximal compact subgroups of $\mathrm{SL}_2(\mathbb{C})$ arise in this way.
4. $\mathrm{GL}_n(\mathbb{C}) = U_n \times \exp(\mathfrak{p})$, for $\mathfrak{p} = \{A \in M_n(\mathbb{C}) : A = \overline{A^t}\}$. This is called the polar decomposition

Proof. 1. This is clear.

2. $\mathrm{SU}_2 = \{\begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} \mid z\bar{z} + w\bar{w} = 1\} \simeq S^3$

3. $\mathrm{SL}_2(\mathbb{C}) = \mathrm{SU}_2 \times \exp(\mathfrak{p})$ for $\mathfrak{p} = \mathfrak{su}_2$ (a special case of the polar decomposition). Hence if we get some element $xy \in \mathrm{SL}_2(\mathbb{C})\backslash SU_n$ (so $0 \neq y \in \exp(\mathfrak{p})$. Then $y = x^{-1}(xy)$ lies in this subgroup. But then y^n is unbounded. Contradiction. So SU_2 is indeed a maximal compact subgroup.

   To see uniqueness, let K be a maximal compact subgroup of $\mathrm{SL}_2(\mathbb{C})$. Pick an Hermitian inner product on $\mathbb{C}^2$. Use an averaging argument, we may assume this inner product is in fact K-invariant. This gives an embedding of K into some copy of $\mathrm{SU}(2)$. By maximality, we know in fact $K = \mathrm{SU}(2)$

□

Exercise. Let $G = \mathrm{GL}_V, d = \dim V \geq 2$, then G acts on $\mathbb{P}V$.

$$\mathrm{stab}_G \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix} = P_d$$

So $\mathbb{P}^{d-1} = \mathrm{GL}_d/P_d$ as a set (in fact as a variety). The problem here is that P_d is not solvable, so has irreducible representations which are not of dimension 1.

Proof. There's nothing to do here. □

Exercise. Let $G = \mathrm{SL}_V = \mathrm{SL}_d, d = \dim V$. Let $B = \{\text{upper triangular matrices}\}$, $\mathcal{B}$ the associated flag variety of G.

1. G acts transitively on $\mathcal{B}$, with the stablizer of $\{\langle e_1\rangle, \cdots, \langle e_1, \cdots, e_d\rangle\}$ is B. This shows $\mathcal{B} \cong G/B$.
2. $\mathcal{B}$ is projective. (Hint: embed $\mathcal{B}$ into Grassmannians, which are projective from Plücker embedding)
3. $\mathrm{Pic}\ \mathcal{B} = \mathbb{Z}^{d-1}$, and every $\mathcal{L}$ is naturally G-equivariant.
4. Show the ample cones are bijective to the very dominant integral weights.
5. Do the same for $\mathrm{Sp}(V)$, where $n = \dim V = 2d, \mathcal{B} = \{F_1 \leq F_2 \cdots \leq F_d | \dim F_i = i, F_i \subseteq F_i^{\perp}\}$.

Proof. 1. This is clear.

2. There is a natural embedding
$$\mathcal{B} \to \mathrm{Gr}(1,d) \times \mathrm{Gr}(2,d) \times \cdots \mathrm{Gr}(d,d)$$
which maps $\langle v_1\rangle, \cdots, \langle v_1, \cdots, v_n\rangle$ to the corresponding elements in the target.

3. This is also a direct consequence of Bruhat decomposition. Recall we have
$$\mathcal{B} = (U^-B/B) \coprod_{\alpha\in\Pi} (U^- s_\alpha B/B) \coprod_{l(w)>1} (U^- wB/B)$$
We dicard all pieces with codimensions ≥ 2, and set $\mathcal{B}' = (U^-B/B) \coprod_{\alpha\in\Pi} (U^- s_\alpha B/B)$. We know we have $\mathrm{Pic}\ (\mathcal{B}) = \mathrm{Pic}\ (\mathcal{B}')$. Note each $U^- s_\alpha B/B$ is an irreducible codimension one closed subvariety inside $\mathcal{B}'$, and they're pairwise disjoint, se we know we have an exact sequence:
$$\mathbb{Z}^{d-1} \to \mathrm{Pic}\ (\mathcal{B}') \to \mathrm{Pic}\ (U^-B/B) \to 0$$
But U^-B/B is an affine $|\Phi^+|$-space, we know it has trivial Picard group. So indeed we have a surjection:
$$\mathbb{Z}^{d-1} \twoheadrightarrow \mathrm{Pic}\ (\mathcal{B}') = \mathrm{Pic}\ (\mathcal{B})$$
On the level of divisors, this is map is just:
$$(m_1, \cdots, m_{d-1}) \mapsto D_{\mathfrak{m}} = m_1 D_{\alpha_1} + m_2 D_{\alpha_2} + \cdots m_{d-1} D_{\alpha_{d-1}}$$
where $D_{\alpha_i} = \overline{U^- s_{\alpha_i} B/B}$ as a Weil divisor of $\mathcal{B}$. Note for each α_i, we have a closed immersion
$$\varphi_i : \mathbb{P}^1 = \{U_{-\alpha_i} B/B \coprod U_{-\alpha_i} s_{\alpha_i} B/B\} \to \mathcal{B}$$
Then $\varphi_i^*(D_{\mathfrak{m}}) = m_i(U^-_{\alpha_i} s_{\alpha_i} B/B)$. Hence the map is also injective. This justifies the Picard is indeed $\mathbb{Z}^{d-1}$.

All these line bundles are in fact G-equivariant. For each $\mathfrak{m}$, we choose $\lambda = -(m_1\omega_1 + \cdots m_{d-1}\omega_{d-1})$, where w_i are the fundamental weights. Consider the B-representation k_λ, and let $\mathcal{L}(\lambda) = G \times^B k_\lambda$ be the corresponding equivariant line bundle on G/B. But then, $\varphi_i^*(\mathcal{L}(\lambda)) = \mathcal{O}(m_i)$, which corresponds to the divisor $m_i(U_{\alpha_i}^- s_{\alpha_i} B/B)$. So indeed $\mathcal{L}(\lambda) = \mathcal{O}(D_\mathfrak{m})$.

4. I can't think of a proof that can simplified from the general case. We still need to use Bruhat decomposition to find the big cell and see which sections can be extended.

5. All previous arguments carry through.

□

Example 1.1. *We carefully walk through the case for $G = \mathrm{SL}_3$. $\mathcal{B} = \{(F_1 < F_2) | \dim(F_i) = i\} = \{(v, v^*) \in \mathbb{P}V \times \mathbb{P}V^* | \langle v^*, v\rangle = 0\}$. Hence there is a canonical map $i : \mathcal{B} \to \mathbb{P}^2 \times \mathbb{P}^2$.*

Lefschetz Hyperplane Theorem tells us that any line bundle on $\mathcal{B}$ is of form:

$$i^*(\mathcal{O}(a) \boxtimes \mathcal{O}(b)) \coloneqq \mathcal{L}_{a,b}$$

Let $\mathcal{I}_\mathcal{B}$ be the ideal of functions vanishing on $\mathcal{B}$ (the defining equations of $\mathcal{B}$):

$$\left\langle s = \sum_{i=0}^{2} x_i y_i \in \Gamma(\mathbb{P}^2 \times \mathbb{P}^2, \mathcal{O}(1) \boxtimes \mathcal{O}(1)) \right\rangle$$

Hence we have $\mathcal{O}_{\mathbb{P}^2\times\mathbb{P}^2} = \mathcal{O} \xrightarrow{s} \mathcal{O}(1) \boxtimes \mathcal{O}(1)$. Dualizing, we get a short exact sequence:

$$0 \to \mathcal{O}(-1) \boxtimes \mathcal{O}(-1) \to \mathcal{O} \to i_*\mathcal{O}_\mathcal{B} \to 0$$

To compute the cohomology of $\mathcal{L}_{a,b}$, we push it forward to $\mathbb{P}^2 \times \mathbb{P}^2$: $i_*\mathcal{L}_{a,b}$. Tensoring the above exact sequence with $i_*\mathcal{L}_{a,b}$, we get:

$$0 \to \mathcal{O}(a-1) \boxtimes \mathcal{O}(b-1) \to \mathcal{O}(a) \boxtimes \mathcal{O}(b) \to i_*\mathcal{L}_{a,b} \to 0$$

We can then use this (and snake lemma) to compute cohomologies of $i_*\mathcal{L}_{a,b}$. In particular: if $a, b \geq 0$,then all higher cohomology vanishes and $\Gamma(\mathcal{B}, \mathcal{L}_{a,b}) = \mathrm{coker}\ (L_{a-1} \otimes L_{b-1} \to L_a \otimes L_b)$, where $L_a = S^a V^* = \Gamma(\mathbb{P}^2, \mathcal{O}(a))$, for $V = k^3$.

Exercise. Assume we're in the setup of SL_3.

1. Check this is correct by decomposing $L_a \otimes L_b$ as $\mathfrak{sl}_3$-representations.

2. Compute $\dim \Gamma(\mathcal{B}, \mathcal{L}_{a,b})$. Check this agrees with Weyl dimension formula.

3. Compute Higher cohomology of $H^i(\mathcal{B}, \mathcal{L}_{a,b})$ for arbitrary a, b.

4. (bonus) There is a fibration $\mathcal{B} \to \mathbb{P}^2$ with fibre $\mathbb{P}^1$. Use this to compute the global section of $\mathcal{L}_{a,b}$.

Proof. 1. Let ω_1, ω_2 be fundamental weights of $\mathfrak{sl}_3$. We know the definining representation is precisely $V(\omega_2)$. From this one can easily see $V(a\omega_1) \simeq L_a = S^a V^*$ and $V(b\omega_2) \simeq L_b = S^b V$, by comparing the highest weight and compute the dimension:

$$L_a \simeq \Gamma(\mathbb{P}^2, \mathcal{O}(a)) \simeq k\langle x_0^i x_1^j x_2^k \mid i+j+k = a\rangle$$

has dimension $\frac{(a+2)(a+1)}{2} = \dim V(a\omega_1)$. Recall by Weyl dimension formula we have

$$\dim V(a\omega_1 + b\omega_2) = \frac{(a+1)(b+1)(a+b+2)}{2}$$

On can thus check

$$\dim V(a\omega_1 + b\omega_2) = \dim V(a\omega_1) \otimes V(b\omega_2) - \dim V((a-1)\omega_1) \otimes V((b-1)\omega_2)$$

and they share the same highest weight vector. So indeed

$$V(a\omega_1 + b\omega_2) = \text{coker } (L_{a_1} \otimes L_{b-1} \to L_a \otimes L_b) = L_{a,b}$$

2. This follows from the first part.

3. Recall if $\mathbb{P}^2 = \mathbb{P}(V)$, then

$$H^i(\mathbb{P}(V), \mathcal{O}(a)) = \begin{cases} S^a V^* = V(a\omega_1), & \text{if } i = 0, a \geq 0; \\ S^{-a-3} V = V((-a-3)\omega_2), & \text{if } i = 2, a \leq -3; \\ 0, & \text{otherwise.} \end{cases}$$

Similarly,

$$H^i(\mathbb{P}(V^*), \mathcal{O}(b)) = \begin{cases} S^b V = V(b\omega_2), & \text{if } i = 0, a \geq 0; \\ S^{-a-3} V^* = V((-b-3)\omega_1), & \text{if } i = 2, a \leq -3; \\ 0, & \text{otherwise.} \end{cases}$$

So by Kunneth formula, we see if $i \neq 0, 2, 4$ then

$$H^i(\mathbb{P}^2 \times \mathbb{P}^2, \mathcal{O}(a) \times \mathcal{O}(b)) = 0$$

for all a, b. Moreover,

$$\begin{aligned} &H^2(\mathbb{P}^2 \times \mathbb{P}^2, \mathcal{O}(a) \boxtimes \mathcal{O}(b)) \\ &= H^0(\mathbb{P}V, \mathcal{O}(a)) \otimes H^2(\mathbb{P}V^*, \mathcal{O}(b)) \bigoplus H^2(\mathbb{P}V, \mathcal{O}(a)) \otimes H^0(\mathbb{P}V^*, \mathcal{O}(b)) \\ &= \begin{cases} V(a\omega_1) \otimes V((-b-3)\omega_1), & \text{if } a \geq 0, b \leq -3; \\ V((-a-3)\omega_2) \otimes V(b\omega_2), & \text{if } a \leq -3, b \geq 0; \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

From the long exact sequence we know: if $a \geq 0, b \leq -2$ then

$$H^1(\mathcal{B}, \mathcal{L}_{a,b}) = \ker(V((a-1)\omega_1) \otimes V((-b-2)\omega_2) \to V(a\omega_1) \otimes V((-b-3)\omega_2))$$

$$H^2(\mathcal{B}, \mathcal{L}_{a,b}) = \text{coker } (V((a-1)\omega_1) \otimes V((-b-2)\omega_2) \to V(a\omega_1) \otimes V((-b-3)\omega_2))$$

and similarly for the other case. Moroever, the degree 3 cohomology followed from Serre's duality.

(But I don't know how to compute these concretely).

□

Exercise. Let $\mathrm{SU}_V \subset \mathrm{SL}_V$ acts on $\mathbb{P}(V)$.

1. Check the above action is transitive, and stab $\begin{bmatrix} 1 \\ 0 \\ \cdots \\ 0 \end{bmatrix} = P_d \cap \mathrm{SU}_V = U_{d-1}$. So, $\mathrm{SU}_V/U_{d-1} \cong \mathrm{SL}_d/P_d = \mathbb{P}V$.
2. for $d = 2$, the right fibration is homotopic to the left one (the Hopf fibration)

$$\begin{array}{ccc} S^1 & \longrightarrow & \mathrm{SU}_2 \\ & & \downarrow \\ & & \mathrm{SU}_2/S_1 = S^2 \end{array} \qquad \begin{array}{ccc} B & \longrightarrow & \mathrm{SL}_2 \\ & & \downarrow \\ & & \mathbb{P}^1 \end{array}$$

3. Let SU_V acts on $\{v \in V | \langle v, v \rangle = 1\} = S^{2d-1}$. This gives a fibration:

$$\begin{array}{ccc} \mathrm{SU}_{d-1} & \longrightarrow & \mathrm{SU}_d \\ & & \downarrow \\ & & S^{2d-1} \end{array}$$

This realizes SU_d as "twisted product" of $S^3, S^5, \cdots, S^{2d-1}$.

Proof. The only content here is that SU_d acts transitively on S^{2d-1}. But this is simply Gram-Schimdt process. □

2 Affine Algebraic Groups

Proposition. *Let $\sigma \in \mathrm{End}(V)$.*

1. *There exists $N, S \in \mathrm{End}(V \otimes \overline{k})$ such that $\sigma = S + N$ for S semisimple, N nilpotent, $SN = NS$.*
2. *These properties characterize S and N uniquely, and $S = p(\sigma)$ for $p(x) \in x\overline{k}[x]$.*
3. *if k is perfect, then $S, N \in \mathrm{End}(V)$.*

Exercise. Let $\sigma \in \mathrm{End}(V)$ be locally finite.

1. Show σ is invertible $\iff$ all eigenvalues of σ are non-zero $\iff$ σ is injective.

2. Deduce the previous propositions still hold if σ is locally finite.

Proof. 1. The later two statements are clearly equivalent (needn't local finiteness). Now assume σ invertible, then if $\sigma(v) = 0 \implies v = \sigma^{-1}\sigma(v) = 0 \implies \sigma$ injective. Conversely, assume $V = \sum W_i$ for each W_i finite-dimensional σ invariant. We set $\tau|_{W_i} = \sigma|_{W_i}^{-1}$, then τ gives the inverse.

2. Note we're forced to have $S|_{W_i} = (\sigma|_{W_i})^{\text{ss}}$, hence uniqueness. We can check this is indeed a linear operator (this can be checked locally). Moreover, σ invertible $\iff$ σ has nonzero eigenvalues $\iff$ S has nonzero eigenvalues $\iff$ S invertible.

□

Exercise. Let G be finite, $\text{char}(k) = p$. Then g semisimple iff $p \nmid \text{ord}(g)$; g unipotent iff $\text{ord}(g) = p^N$ for some N.

Proof. We study how g acts on $k[G] = \oplus V_k$, where each V_k has basis $\{g^i h\}$ for some h. Note g acts on V_k as matrix

$$T = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & \cdots & \cdots & 1 \\ 1 & & & 0 \end{pmatrix}$$

It has characteristic polynomial $T^d - 1$, for $d = \text{ord}(g)$. Now if $p \nmid d \implies T^d - 1$ has distinct roots, so indeed g is semisimple. Conversely, assume $d = p^k l$ for $p \nmid l$. Then $T^d - 1 = (T^l - 1)^{p^k}$. Assume g is semisimple, then T is diagonalizable, so its minimal polynomial has no repeated roots. Hence the minimal polynomial divides $T^l - 1$. But $T^l - 1 \neq 0$, contradiciton. So g is not semisimple. The discussion for unipotent elements is similar. □

Exercise. Let $g = su \in G$ be Jordan decomposition. Then:

$$Z_G(g) = Z_{Z_G(s)}(u) = Z_G(s) \cap Z_G(u).$$

Proof. The second equality is tautology. Now let $V = k[G]$, and $G \hookrightarrow \text{GL}(V)$ is a locally finite representation. Note $Z_G(g) \supset Z_G(s) \cap Z_G(u)$ is obvious. Coversely, if $x \in Z_G(g)$. Write $V = \cup W_i$ as finite-dimensional G-invariant spaces. On each W_i, $s = p_i(g)$ for some polynomial p_i, so $xs = sx$ on each $W_i \iff sx = xs$ on $V \iff x \in Z_G(s)$. But then $x(su) = (su)x \iff s(xu) = s(ux) \iff xu = ux$ since s is an invertible element in $\text{GL}(V)$. So $x \in Z_G(u)$. □

3 Quotients

Definition. Let G acts on X freely (so every orbit is isomorphic to G). We call X a **principal G-bundle** or **princpal G-torsor** on Y in Zariski topology, if there is a map $p : X \to Y$ such that:

1. if $U \subset Y$ Zariski open, $\Gamma(U, \mathcal{O}) = \Gamma(p^{-1}U, \mathcal{O})^G$

2. for any $y \in Y$, there is a Zariski affine open nieghborhood U, such that $p^{-1}U \cong U \times G$ equivariantly.

3. each fibre is isomorphic to G as an algberaic variety.

4. $p : X \to Y$ is flat, in particular p is open and affine (inverse image of affine open is affine).

Exercise. Show that if the above properties hold on some affine cover, then they hold on all affine opens.

Proof. The conditions 2,3,4 evidently can be checked on an affine cover. For condition 1, let U be an Zariski open set with $U = \bigcup U_i$, for U_i open affines. So

$$\begin{aligned}\Gamma(p^{-1}U, \mathcal{O})^G &= \{(f_i \in \Gamma(p^{-1}U_i, \mathcal{O})^G)_{i\in I} \mid f_i|_{p^{-1}U_i \cap p^{-1}U_j} = f_j|_{p^{-1}U_i \cap p^{-1}U_j}\} \\ &= \{(f_i \in \Gamma(U_i, \mathcal{O}))_{i\in I} \mid f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}\} \\ &= \Gamma(U, \mathcal{O})\end{aligned}$$

as required. □

Exercise. 1. Let $\mathrm{PGL}_n = \mathrm{GL}_n/\mathbb{G}_m$ is an affine algebraic group by writing its defining equations. Note $\mathrm{PGL}_n \subset (\mathrm{Mat}_n \backslash \{0\})/\mathbb{G}_m = \mathbb{P}^{n^2-1}$ is a quasi-projective variety given by nonvanishing determinant. Also, show that the orbit construction is really a quotient.

2. Show $\mathrm{PSL}_n = \mathrm{SL}_n/\mu_n$ is an affine algebraic group. So we have:

$$\begin{array}{ccccc} \mu_n & \hookrightarrow & \mathbb{G}_m & \xrightarrow{det} & \mathbb{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{SL}_n & \hookrightarrow & \mathrm{GL}_n & \xrightarrow{det} & \mathbb{G}_m \\ \downarrow & & \downarrow & & \\ \mathrm{PSL}_n & \longrightarrow & \mathrm{PGL}_n & & \end{array}$$

Show that $\mathrm{PSL}_n \cong \mathrm{PGL}_n$.

Proof. 1. Let $G = \mathbb{G}_m$. Note $\mathrm{GL}_n = \mathrm{Spec}\ k[x_{ij}, \det(x_{ij})^{-1}] = \mathrm{Spec}\ A$. We easily see that $A^G = k[\frac{x^{\mathbf{i}}}{\det(x_{ij})}]$, where $x^{\mathbf{i}}$ runs through all degree n monomials in $k[x_{ij}]$. We need to check the morphism $p : \mathrm{Spec}\ A \to \mathrm{Spec}\ A^G$ is indeed a flat G-quotient. But note $\mathrm{Spec}\ A^G$ closed subscheme of the image of $k[x^{\mathbf{i}}] \hookrightarrow \mathbb{A}^N$ cut out by the hyperplane $\det = 1$, so $\mathrm{Spec}\ A^G$ is regular. But it is also clear that the projection $p : \mathrm{Spec}\ A \to \mathrm{Spec}\ A^G$ has fibre isomorphis to G, and $\dim A = \dim A^G + 1 = \dim A^G + \dim G$. So by Miracle flatness, we know p is indeed a flat G-quotient.

2. We know μ_n is a finite group, so SL_n/μ_n is indeed affine from lecture. Note $\mathrm{SL}_n = \mathrm{Spec}\ k[x_{ij}]/(\det(x_{ij}) - 1) = \mathrm{Spec}\ B$. So $\mathrm{PSL}_n = \mathrm{Spec}\ B^{\mu_n} = \mathrm{Spec}\ k[x^{\mathbf{i}}]/(\det(x_{ij}) - 1)$. Hence to show $\mathrm{PSL}_n \simeq \mathrm{PGL}_n$, we only need to show the natural map

$$k[x^{\mathbf{i}}/\det(x_{ij})] \to k[x^{\mathbf{i}}]/(\det(x_{ij}) - 1)$$

is an isomorphism. But this is implicitly done in the geometric picture in part 1.

□

Exercise. Show that if $G \times_Y p^{-1}(U) \cong p^{-1}(U) \times_Y p^{-1}(U)$ for U in some fixed affine cover of Y. Then $G \times_Y X \cong X \times_Y X$ is indeed an isomorphism.

Proof. We are given isomorphisms $f_{U_i} : G \times_Y p^{-1}(U_i) \to p^{-1}(U_i) \times_Y p^{-1}(U_i)$. We know $f_{U_i}|_{U_i \cap U_j} = f_{U_j}|_{U_i \cap U_j}$, since they are given by the same formula. We can hence glue $f_{U_i}^{-1}$ to obtain the inverse of the morphism $(g, x) \mapsto (gx, x)$. □

Exercise. $G \times X \cong X \times_Y X$ can be checked pointwise. Moreover, $X(A)/G(A) \hookrightarrow Y(A)$.

Proof. The first statement is simply the functor of point perspective. For the second statement, we have

$$p : X(A) \to Y(A)$$

and bijeciton

$$G(A) \times_{Y(A)} X(A) \to X(A) \times_{Y(A)} X(A) \quad (g, x) \mapsto (gx, x)$$

Note for $x \in X(A)$ and $g \in G(A)$, $p(gx) = p(x)$ since the bijection is well-defined. So indeed we have an induced map

$$\bar{p} : X(A)/G(A) \to Y(A)$$

To check this injective, if $\bar{p}(x) = \bar{p}(y)$, then $(x, y) \in X(A) \times_{Y(A)} X(A) \implies \exists (g, t) \in G(A) \times X(A), gt = x, t = y \implies x = y \bmod G(A)$. □

4 Flag Varieties

Theorem (Borel Fixed Point Theorem, BFPT)**.** *Let G be a connected solvable affine algebraic group acting on a proper variety X, then G has a fixed point on X.*

Exercise. (Alternative Proof for BFPT)
Claim: If G is connected reduced affine algebraic group of dimension 1, then $G = \mathbb{G}_a, \mathbb{G}_m$.
Claim: If G reduced solvable connected affine aglebraic group, there exists a normal series $1 < H_1 < H_2 < \cdots < H_r = G$ with $H_i \lhd H_{i+1}$ and each quotient has dimension 1.
Claim: Let $G = \mathbb{G}_a$ or $\mathbb{G}_m$, acting on proper X. Let $x \in X$. If $G \cdot x \neq x$, then we get a map

$$\begin{array}{ccccc} G & \longrightarrow & G \cdot x & \hookrightarrow & X \\ & & \downarrow & \nearrow & \\ & & \mathbb{P}^1 & & \end{array}$$

valuation criterion for properness shows that extends to a map $\varphi : \mathbb{P}^1 \to X$. Show $\varphi(\infty)$ is fixed by G
Claim: Use these claims and inuciton to finish the proof.

Exercise. (Burnside). Let Γ be a (semi)group acting faithfully, irreducibly on V for V finite dimensional vector space. Consider $\#\{\mathrm{tr}(g)|g \in \Gamma\} = r < \infty$. Show that

$$n^2 \leq |\Gamma| \leq r^{n^2}$$

Use this to show the only irreducible representation for a unipotent group is the trivial representation.

Proof. We know the image of $k\Gamma$ spans $\mathrm{End}(V)$ by Weddernburn's theorem. Hence $n^2 \leq |\Gamma|$. On the other hand, pick a basis x_{ij} for $\mathrm{End}(V)$. Consider the map:

$$\gamma \in \Gamma \mapsto (\mathrm{tr}(\gamma x_{ij}))_{i,j}$$

This is injective, hence the right inequality. Now assume Γ is unipotent, then $r = 1$ and this forces $n = 1$ and Γ to be the trivial group. □

Exercise. Distinct characters $\chi_j : G \to \mathbb{G}_m$ are linearly independent. In particular, characters span $k[G]$ if and only if they form a basis.

Proof. Let $\chi_1, \cdots, \chi_n$ be linearly dependent for n minimal. Clearly $n \geq 2$. Suppose $c_1\chi_1 + \cdots c_n\chi_n = 0$. So we have

$$c_1\chi_1(g) + \cdots c_n\chi_n(g) = 0$$

for each $g \in G$. Choose g_0 such that $\chi_1(g_0) \neq \chi_n(g_0)$. Then we have:

$$c_1\chi_1(g_0g) + \cdots c_n\chi_n(g_0g) = c_1\chi_1(g_0)\chi_1(g) + \cdots c_n\chi_n(g_0)\chi_n(g) = 0$$

and

$$c_1\chi_n(g_0)\chi_1(g) + \cdots c_n\chi_n(g_0)\chi_n(g) = 0$$

Substract these two expression, we get:

$$c_1(\chi_1(g_0) - \chi_n(g_0))\chi_1(g) + \cdots c_{n-1}(\chi_{n-1}(g_0) - \chi_n(g_0))\chi_{n-1}(g) = 0$$

This gives a linearly dependency of $\chi_1, \cdots, \chi_{n-1}$, contradicting the minimality of n. □

Exercise.

1. G acts on $\mathfrak{g}$ via Lie algebra automorphism: $[\mathrm{Ad}(g)x, \mathrm{Ad}(g)y] = \mathrm{Ad}(g)[x, y]$
   In fact, $G^\circ/Z_G = \mathrm{Aut}^0(\mathfrak{g})$.
2. Hence $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$

Proof.

1. Recall we have $\varphi_g : G \to G, x \mapsto gxg^{-1}$. We differentiate at identity:

   $$\mathrm{Ad}(g) = d\varphi_g|_e \colon \mathfrak{g} \to \mathfrak{g}$$

   If $x \in \mathfrak{g}$ is a derivation $k[G] \to k$, then $\mathrm{Ad}(g)$ acts as:

   $$\begin{array}{ccc} k[G] & \xrightarrow{\varphi_g^*} & k[G] \\ & \searrow^{\mathrm{Ad}(g)(x)} & \downarrow{\scriptstyle x} \\ & & k \end{array}$$

   So $\mathrm{Ad}(g)(x) = x \circ \varphi_g^*$. So $\mathrm{Ad}(g)[x, y] = [\mathrm{Ad}(g)x, \mathrm{Ad}(g)y]$ is equivalent to $[x \circ \varphi_g^*, y \circ \varphi_g^*] = [x, y] \circ \varphi_g^*$, but this is clear since φ_g^* is a ring isomorphism.

2. If $\mathrm{Ad}(t)x = \alpha(t)x, \mathrm{Ad}(t)y = \beta(t)y$, then

$$\mathrm{Ad}(t)[x,y] = [\mathrm{Ad}(t)x, \mathrm{Ad}(t)y] = [\alpha(t)x, \beta(t)y] = (\alpha+\beta)(t)[x,y]$$

□

Exercise. For $G = \mathrm{SL}_2$, show $X^*(T) = Y$, where Y is weight lattice for the associated Lie algebra; for $G = \mathrm{PGL}_2$, show $X^*(T) = 2Y \subsetneq Y$.

Proof. Let $G = \mathrm{SL}_2$. Then we've seen in lecture $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. So

$$Y = \{f \in \mathfrak{t}^* \mid f(H) \in \mathbb{Z}\}$$

On the other hand, we have

$$X^*(T) = \{f : T \to \mathbb{G}_m\} = \{f_n : \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \mapsto t^n\} \simeq \{df_n|_1 \colon H \mapsto n\} = Y$$

On the other hand, for $G = \mathrm{PGL}_2$ we can parametrize the torus:

$$\mu^\vee : \mathbb{G}_m \to T, t \mapsto \begin{pmatrix} t & \\ & 1 \end{pmatrix}$$

where $T \simeq \begin{pmatrix} * & \\ & * \end{pmatrix} / \begin{pmatrix} a & \\ & a \end{pmatrix}$ and hence $\mathfrak{t} \simeq \begin{pmatrix} * & \\ & * \end{pmatrix} / \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$.

Set $H' = d\mu^\vee|_1(1) = \begin{pmatrix} 1 & \\ & 0 \end{pmatrix}$. One sees $H = 2H'$ fits into the $\mathfrak{sl}_2$-triple. So

$$X^*(T) = \{f_n : \begin{pmatrix} t & \\ & 1 \end{pmatrix} \mapsto t^n\} \simeq \{df_n|_1 \colon \begin{pmatrix} 1 & \\ & 0 \end{pmatrix} \mapsto n\}$$

On the other hand, we know

$$Y = \{f \in \mathfrak{t}^* \mid f(H) \in \mathbb{Z}\} = \{f \in \mathfrak{t}^* \mid f\begin{pmatrix} 1 & \\ & 0 \end{pmatrix} \in \frac{1}{2}\mathbb{Z}\}$$

So indeed $X^*(T) = 2Y$. □

Exercise. Prove Bruhat Decomposition for classical groups.

Proof. We need the following general lemma.

Lemma. *Let G be any semisimple group with Weyl group W. Let $w \in W, U(w) = U \cap wU^-w^{-1}, U(w)' = U \cap wUw^{-1}$. Fix a choice of representative n_w of $w \in W$. Then each element in BwB can be uniquely written as un_wb for $u \in U(w), b \in B$.*

Proof. (of lemma) Note that the Lie algebra for $U(w)$ is $\bigoplus_{\alpha>0,w(\alpha)<0}\mathfrak{g}_\alpha$, and the Lie algebra for $U(w)'$ is $\bigoplus_{\alpha>0,w(\alpha)>0}\mathfrak{g}_\alpha$. So it follows that

$$U = U(w)U(w)'$$

We also know $n_w T = T n_w$. So we compute:

$$BwB = U(w)U(w)'Tn_wB = U(w)U(w)'n_wTB = U(w)U(w)'n_wB = U(w)n_wB$$

To see uniqueness, assume $n_w = un_wb \implies n_w^{-1}u^{-1}n_w = b \in U^-$. So $b = 1$ and $u = 1$. □

1. $G = \mathrm{SL}_n$. Then $N(T)$ consists of monomial matrices in SL_n. Given $g \in G$. We can choose $b \in B$ such that each row of $b \cdot g^{-1}$ has leading zeroes as many as possible. We can the apply $w \in N(T)$ such that $wbg^{-1} \in B \implies g \in Bwb$. Hence the double cosets BwB indeed covers G. To see these double cosets are disjoint, suppose $w' = b'wb$ for some $b', b \in B$. We have:
$$w'b^{-1}w^{-1} = b'$$
Suppose w, w' correspond to permutations σ, σ'. Apply the above equation to $e_{\sigma(j)}$, we see the left hand side has support
$$\{\sigma'(i) \mid b_{ij}^{-1} \neq 0\}$$
So:
$$b_{ij}^{-1} \neq 0 \implies \sigma'(i) \leq \sigma(j)$$
In particular $b_{jj}^{-1} \neq 0 \implies \sigma'(j) \leq \sigma(j)$ for each j. Hence $\sigma = \sigma'$.

2. $G = \mathrm{Sp}_{2n}$. Set $J_n = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ & \cdots & \cdots & \cdots & \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}$, and $M = \begin{pmatrix} 0 & J_n \\ -J_n & 0 \end{pmatrix}$. Let $\varphi : A \mapsto M^{-1} \cdot {}^tA^{-1} \cdot M$ be an automorphism of SL_{2n}. Then $\mathrm{Sp}_{2n} = \{A \in \mathrm{SL}_{2n} \mid \varphi(A) = A\}$. We observe that φ preserves $T, B, N(T), U, B^-, U^-$ of SL_{2n}. Then $\mathrm{Sp}_{2n} = \{A \in \mathrm{SL}_{2n} \mid \varphi(A) = A\}$. Now if $g \in \mathrm{Sp}_{2n}$, we write $g = un_wb$ as in the lemma. We know:
$$g = \varphi(g) = \varphi(u)\varphi(n_w)\varphi(b)$$
By uniqueness of the lemma, we know
$$\varphi(u) = u, \varphi(n_w) = n_wt, \varphi(b) = t^{-1}b$$
for some $t \in T$. We adjust n_w such that $t = 1$. Then indeed each u, n_w, b indeed lies in Sp_{2n}. This justifies the existence. The uniqueness is clear from the uniqueness for the case of SL_n.

3. $G = \mathrm{SO}_n$. The idea is very similar. It suffices to find a bilinear form M compatible with $T, B, N(T), U, B^-, U^-$ of SO_n. Indeed, for $n = 2m$, choose $M = \begin{pmatrix} 0 & J_n \\ J_n & 0 \end{pmatrix}$; for $n - 2m + 1$ choose $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & J_n \\ 0 & J_n & 0 \end{pmatrix}$

□

5 Representation of Semisimple Groups

Exercise. There is no section $\mathrm{PGL}_2 \to \mathrm{GL}_2$.

Proof. If so, then $\mathrm{GL}_2 = k \rtimes \mathrm{PGL}_2 \simeq k \times \mathrm{PGL}_2$, since the copy of k lies in the center of GL_2. We can take the commutator bracket:

$$\mathrm{SL}_2 = [\mathrm{PGL}_2, \mathrm{PGL}_2] = \mathrm{PSL}_2.$$

□

Exercise. Prove Frobenius Reciprocity. Let V be any G-representation. Then:

$$\mathrm{Hom}_G(V, \Gamma(G/B, \mathcal{L}_\lambda)) \simeq \mathrm{Hom}_B(V, k_\lambda)$$

Proof. We construct the map on both sides. There is clearly a map

$$\varphi : \mathrm{Hom}_G(V, \Gamma(G/B, \mathcal{L}_\lambda)) \to \mathrm{Hom}_B(V, k_\lambda)$$

defined by

$$f \mapsto (v \mapsto f(v)|_B)$$

Conversely, ther is a map

$$\psi : \mathrm{Hom}_B(V, k_\lambda) \to \mathrm{Hom}_G(V, \Gamma(G/B, \mathcal{L}_\lambda))$$

defined by

$$h \mapsto (v \mapsto (\psi_{h,v} : G \to k_\lambda, g \mapsto h(g^{-1}v))$$

We need to show this is well-defined:

$$\psi_{h,v}(gb) = h(b^{-1}g^{-1}v) = b^{-1}h(g^{-1}v) = \lambda(b)^{-1}\psi_{h,v}(g)$$

Moreover, this is clearly G-equivariant:

$$g \cdot \psi_{h,v}(x) = \psi_{h,v}(g^{-1}x) = h(x^{-1}gv) = \psi_{h,gv}(x) \implies g \cdot \psi_{h,v} = \psi_{h,gv}$$

One can then show φ, ψ are inverse to each other. □

Exercise. Show $L(\lambda)^* = L(-w_0\lambda)$

Proof. Note $L(\lambda)^{U^-} = k_\lambda$ from lecture. So we have

$$\mathrm{Hom}_B(L(\lambda), k_\lambda) \neq 0$$

It follows that

$$\mathrm{Hom}_B(k_{-\lambda}, L(\lambda)^*) \neq 0$$

Now consider the group automorphism $g \mapsto w_0 g w_0^{-1}$, this maps B to B^-,and under this twisted action $k_{-\lambda}$ is has weight $-w_0\lambda$. On the other hand, since $L(\lambda)^*$ is a representation of G, and we know its character is stable under Weyl group element, so it remains to be $L(\lambda)^*$ is the twisted action. Hence we in fact have

$$\mathrm{Hom}_{B^-}(k_{-w_0\lambda}, L(\lambda)^*) \neq 0$$

Hence $(L(\lambda)^*)^{U^-}$ has a weight $-w_0\lambda$. But $L(\lambda)^*$ is irreducible, it is forced to be $L(-w_0\lambda)$. □

Exercise. 1. If $\lambda \in X^+$, then $\mathcal{L}_\lambda$ is ample on G/P_λ

2. If $\lambda \in Y^+$, then $H^i(G/P_\lambda, \mathcal{L}_\lambda) = H^i(G/B, \mathcal{L}_\lambda)$.

Proof. 1. Now we have closed immersion $G/P_\lambda = G \cdot l \hookrightarrow \mathbb{P}V$ for some finite-dimensional G-reps V, where $l = kv_\lambda$ such that $\mathrm{stab}_G(l) = P_\lambda$. Now assume P_λ acts on l via character μ. In particular, $[P_\lambda, P_\lambda]$ acts trivially on v_λ. So if $\alpha \in \Phi_\lambda$, then $H_\alpha v_\lambda = 0 \implies \langle \mu, \alpha^\vee \rangle = 0$. We choose m large enough such that $-\mu + m\lambda \in X^+$, so that we can find irreduicble representation W of G such that $W^U = kw$ has weight $-\mu + m\lambda$.

If $\alpha \in \Phi_\lambda$, then $\langle \lambda, \alpha^\vee \rangle = 0$. Moreover, $\langle -\mu + m\lambda, \alpha^\vee \rangle = 0$ by previous discussion. Hence in fact P_λ fixes the line kw. We can now proceed as we've seen in the lecture. We have the embedding:

$$G/P_\lambda \hookrightarrow \mathbb{P}(V \otimes W)$$

and indeed $\mathcal{L}(m\lambda) = i^*(\mathcal{O}_{\mathbb{P}(V \otimes W)}(1))$.

2. Consider the map

$$f : G/B \to G/P$$

We've seen in lecture that this is a locally trivial P/B-fibration with projective fibres. This implies that f is a rational morphism. Then we can compute the cohomology of line bundles via pullback.

□

Exercise. Let $G = \mathrm{Sp}(2n)$ be the group of automorphisms preserving a symplectic form. Let $\mathcal{B} = \{F_1 < F_2 < \cdots < F_n \mid \dim F_i = i, F_i \leq F_i^\perp\}$ be its associated flag variety.

1. Show $\mathcal{B}$ is projective and G acts on it transitively.

2. Pick any closed point $F \in \mathcal{B}$, the stablizer $\mathrm{stab}_G(F)$ is a Borel subgroup of G. Choose $T \subset B$ be a maximal torus.

3. Let $\mathfrak{g} = \mathrm{Lie}(G)$ be its Lie algebra. Compute the root space decomposition of $\mathfrak{g}$.

4. Compute the root datum$\{X^*(T), \Phi, X_*(T), \Phi^\vee\}$

5. Compute the Weyl group.

6. State and prove the Bruhat decomposition for $\mathrm{Sp}(2n)$

7. For each $\lambda \in X^*(T)$, describe the line bundle $\mathcal{L}_\lambda$ at each point $F \in \mathcal{B}$

   For any subset $I = \{i_1 < i_2 < \cdots < i_r\} \subset \{1, 2, \cdots, n\}$, let $\mathcal{P}_I = \{F_{i_1} < F_{i_2} < \cdots < F_{i_r} \mid \dim F_{i_j} = i_j, F_{i_j} \leq F_{i_j}^\perp\}$. Let $\pi_I : \mathcal{B} \to \mathcal{P}_I$ be the natural projection. Let $F_I = \pi_I(F)$

8. Describe $\mathrm{stab}_G(F_I)$. The fibre $\pi_I^{-1}(F_I)$ is a flag variety of certain affine algebraic group. Identify this fibre.

9. Let Z_G be its centre. Show this is an affine algebraic group.

10. Prove $\mathrm{PSp}(2n) = G/Z_G$ is an affine algebraic group. Compute its root datum.

11. Do the same computations for $\mathrm{SL}_n, \mathrm{SO}_{2n+1}, \mathrm{SO}_{2n}$.

Proof. We assume the symplectic form is given by $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. So if we choose basis $\{h_1, h_2, \cdots, h_n, f_1, \cdots, f_n\}$, then the symplectic form is given by $\omega(h_i, h_j) = \omega(f_i, f_j) = 0, \omega(h_i, f_j) = \delta_{ij} = -\omega(f_j, h_i)$.

1. We have a closed embedding:

$$\mathcal{B} \to \mathrm{Gr}(1, 2n) \times \mathrm{Gr}(2, 2n) \times \cdots \mathrm{Gr}(n, 2n)$$

   This is closed because all the defining properties are closed properties, since they can be written as certain rank conditions. To see G acts transitively, we first note that each $F = \{F_1 < F_2 < \cdots < F_n\} \in \mathcal{B}$ can be uniquely extended to a symplectic complete flag in k^{2n}:

$$F_1 < F_2 < \cdots < F_n < F_{n-1}^{\perp} < F_{n-2}^{\perp} < \cdots F_1^{\perp} < k^{2n}$$

   We choose any flag $F = \{F_1 < \cdots < F_n\} \in \mathcal{B}$, where $F_i = \langle x_1, \cdots, x_i \rangle$. We now want to construct $y_1, \cdots, y_n$ such that $F_{n-k}^{\perp} = \langle x_1, \cdots, x_n, y_{n-k+1}, \cdots, y_n \rangle$, $\omega(y_i, y_j) = 0, \omega(x_i, y_j) = \delta_{ij} = -\omega(y_j, x_i)$. In fact we can just choose y_j backwards inductively (start from y_n) and apply Gram-Schmidt process.

   Now there exists $g \in \mathrm{GL}_{2n}$ such that $g(h_i) = x_i, g(f_j) = y_j$. Hence in particular g preserves ω so $g \in \mathrm{Sp}(2n)$. Moreover, g maps the standard flag $\{\langle h_1 \rangle < \langle h_1, h_2 \rangle < \cdots < \langle h_1, h_2, \cdots, h_n \rangle\}$ to F.

2. Let's choose F to be the standard flag. Let $B = \mathrm{stab}_G(F)$. Note this forces each $b \in B$ to have form $\begin{pmatrix} A & D \\ 0 & C \end{pmatrix}$ where A is upper triangular. But we must have

$$\begin{pmatrix} A^t & \\ D^t & C^t \end{pmatrix} \cdot \begin{pmatrix} & I \\ -I & \end{pmatrix} \cdot \begin{pmatrix} A & D \\ & C \end{pmatrix} = \begin{pmatrix} & I \\ -I & \end{pmatrix}$$

   It follows that $C = (A^t)^{-1}$. $D^t C = C^t D$. In particular C is lower triangle. But we can rearrange the basis to make the matrix upper triangule. So B is indeed solvable. On the other hand, we know $\mathcal{B} \simeq G/B$ is projective. So B contains some Borel subgroup. Hence B is in fact a Borel subgroup. One also sees the diagonal elements in B form a maximal torus. Indeed, one can easily see that all the diagonal 1-elements in B form the unipotent radical of B, so B/U has dimension n. But $T = \{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1})\}$ also has dimension n and is a torus. So it is indeed a maximal torus.

3. The Lie algebra

$$\mathfrak{g} = \mathfrak{sp}(2n) = \{g \in \mathfrak{gl}_n \mid \omega(gx, y) + \omega(x, gy) = 0, \forall x, y \in k^{2n}\}$$

it consists of matrices of form $\begin{pmatrix} A & C \\ D & -A^t \end{pmatrix}$, where C, D are symmetric matrices. Hence $\mathfrak{g}$ has a basis of form

$$\{E_{ij} - E_{n+j,n+i}\} \cup \{E_{i,n+j} + E_{j,n+i} \mid i \leq j\} \cup \{E_{i+n,j} + E_{j+n,i} \mid i \leq j\}$$

Now if $t = \mathrm{diag}(t_1, \cdots, t_n^{-1})$, we compute:

$$t(E_{ij} - E_{n+j,n+i})t^{-1} = \frac{t_i}{t_j}(E_{ij} - E_{n+j,n+i})$$

$$t(E_{i,n+j} + E_{j,n+i})t^{-1} = (t_i t_j)(E_{i,n+j} + E_{j,n+i})$$

$$t(E_{i+n,j} + E_{j+n,i})t^{-1} = \frac{1}{t_i t_j}(E_{i+n,j} + E_{j+n,i})$$

Hence $X^*(T) = \mathrm{Hom}(T, \mathbb{G}_m) \simeq \mathbb{Z}^n$, Then

$$\Phi = \{t \mapsto \frac{t_i}{t_j}\} \cup \{t \mapsto t_i t_j\} \cup \{t \mapsto \frac{1}{t_i t_j}\} \simeq \{e_i - e_j \mid i \neq j\} \cup \{e_i + e_j, -e_i - e_j\} \subset \mathbb{Z}^n$$

where $e_i : t \mapsto t_i$ are the standard basis for $\mathbb{Z}^n$. So

$$\mathfrak{g} = \mathfrak{t} \bigoplus_{i \neq j} \mathfrak{g}_{e_i - e_j} \bigoplus_{i \leq j} \mathfrak{g}_{e_i + e_j} \bigoplus_{i \leq j} \mathfrak{g}_{-e_i - e_j}$$

is the root space decomposition of $\mathfrak{g}$.

4. We've seen $X^*(T) = \mathbb{Z}\langle e_1, \cdots, e_n \rangle$ and $\Phi = \{e_i - e_j \mid i \neq j\} \cup \{e_i + e_j, -e_i - e_j\}$. Now if we set

$$e_i^\vee : \mathbb{G}_m \to T, u \mapsto \mathrm{diag}(1, \cdots, u, \cdots, u^{-1}, \cdots, 1)$$

where u, u^{-1} are at $i, n+i$ places. Then clearly $\langle e_i, e_j^\vee \rangle = \delta_{ij}$, and $X_*(T) = \mathbb{Z}\langle e_1^\vee, \cdots e_n^\vee \rangle$. To describe coroots, we need to see the copies of SL_2 sitting inside G corresponding to Φ:

$$\varphi_{e_i - e_j} : \begin{pmatrix} u & \\ & u^{-1} \end{pmatrix} \mapsto (e_i^\vee - e_j^\vee)(u)$$

So indeed $(e_i - e_j)^\vee = e_i^\vee - e_j^\vee$. Similarly, for $i \neq j$, we have:

$$(e_i + e_j)^\vee = e_i^\vee + e_j^\vee \quad (-e_i - e_j)^\vee = -e_i^\vee - e_j^\vee$$

But if $i = j$, then

$$\varphi_{2e_i} \begin{pmatrix} u & \\ & u^{-1} \end{pmatrix} = e_i^\vee(u)$$

Hence $(2e_i)^\vee = e_i^\vee$. In summary:

$$\Phi = \{\pm e_i \pm e_j \mid i < j\} \cup \{\pm 2e_i\}$$

$$\Phi^\vee = \{\pm e_i^\vee \pm e_j^\vee \mid i < j\} \cup \{\pm e_i^\vee\}$$

One sees that $Y = X^*(T)$ and $\mathbb{Z}\Phi^\vee = X_*(T)$. Reconfirm that $\mathrm{Sp}(2n)$ is simply connected.

5. We want to describe $N(T)$. Suppose $g = (a_{ij}) \in N(T)$. We know

$$g \cdot \mathrm{diag}(\lambda_1, \cdots, \lambda_n, \lambda_1^{-1}, \cdots, \lambda_n^{-1}) = \mathrm{diag}(\mu_1, \cdots, \mu_n, \mu_1^{-1}, \cdots, \mu_n^{-1}) \cdot g$$

For simplicity we take convention $\lambda_{n+i} \coloneqq \lambda_i^{-1}$, $\lambda_{i+2n} = \lambda_i$ and similalry for μ. So if $a_{ij} \neq 0$, then $\lambda_j = \mu_i$. Now if there exists $j_1 \neq j_2$ such that a_{i,j_1}, a_{i,j_2} both nonzeroes, then $\lambda_{j_1} = \mu_i = \lambda_{j_2}$ is by force, contradiction. This forces g must be a monoidal matrix. Moreover, $a_{ij} \neq 0 \iff \lambda_j = \mu_i \iff \lambda_{j+n} = \mu_{i+n} \iff a_{i+n,j+n} \neq 0$, where the index are taken to be modulo $2n$. Hence we have

$$N(T) = \{(a_{ij}) \in G \text{ monoidal} \mid a_{ij} \neq 0 \iff a_{i+n,j+n} \neq 0\}$$

From this we see $W = N(T)/T \simeq \{\sigma \in S_{2n} \mid \sigma(i) + n = \sigma(i+n)\}$

6. Consider a pair of symplectic flags F, F'. Fix i and look at $F'_{i-1} + F'_i \cap F_j$ for j increases. There will be precisely one such j where the dimension increases. Then the permutation sends j to i.

7. This can be described in general. We know

$$\mathcal{L}_\lambda|_{gB/B} = \{f : gB \to k_\lambda \mid f(gb) = \lambda(b)^{-1} f(g)\} \simeq \{f : g \to k_\lambda\}$$

This is a gBg^{-1} representation inflated from the representation λ of the twisted torus gTg^{-1}.

More concretely, let $F = (F_1 < F_2 < \cdots F_n)$ be a flag in G/B. Note Then:

$$\mathcal{L}_{\lambda,F} = F_1^{\otimes \langle \lambda, e_1^\vee \rangle} \otimes (F_2/F_1)^{\otimes \langle \lambda, e_2^\vee \rangle} \otimes \cdots \otimes (F_n/F_{n-1})^{\otimes \langle \lambda, e_n^\vee \rangle}$$

where each $V^{\otimes -n} = (V^*)^{\otimes n}$

8. Let F be the standard flag

$$P_I = \mathrm{stab}_G(F_I) \simeq L \ltimes U = (\mathrm{GL}(i_1) \times \mathrm{GL}(i_2 - i_1) \times \cdots \times \mathrm{GL}(i_r - i_{r-1}) \times \mathrm{Sp}(2(n - i_r))) \ltimes U$$

where U is the unipotent radical for P_I.

Clearly G acts transitively on $\mathcal{P}_I$, since G actually acts transitively on $\mathcal{B}$. But this action has stablier $\mathrm{stab}_G(F_I)$. So in fact we have

$$\mathcal{P}_I \simeq G/\mathrm{stab}_G(F_I)$$

Hence the projection map can be identified with

$$G/B \to G/\mathrm{stab}_G(F_I)$$

So $\pi_I^{-1}(F_I) \simeq \mathrm{stab}_G(F_I)/B$. We want to show this is indeed the flag variety of $P_I = \mathrm{stab}_G(F_I)$. This is equivalent to show B is maximal solvable subgroup of P_I. But this is clear since B is actually maximal inside G.

9. Suppose $g \in Z_G$. We know all Borel are conjugates, and g is stable under conjugation. So we know g lies in all Borels. In partciular $g \in B \cap B^- = T$. One can then brute forcely check $g = \pm I_{2n}$. Hence $Z_G = \{\pm I_{2n}\}$ is a finite group, in particular affine.

10. Since Z_G is finite, so G/Z_G is well-defined and remains to be an affine algebraic group. It remains to describe the root datum. Now the torus is $T_{\mathrm{PSp}} = T/\pm 1$ which remains to be a n-dimensional torus. We still set $e_i, e_i^\vee$ as before, and

$$\Phi = \{\pm e_i \pm e_j \mid i < j\} \cup \{\pm 2e_i\}, \quad \Phi^\vee = \{\pm e_i^\vee \pm e_j^\vee \mid i < j\} \cup \{\pm e_i^\vee\}$$

We see all elements in Φ indeed descends to $T/\pm 1 \to \mathbb{G}_m$. In fact:

$$X^*(T_{\mathrm{PSp}}) = \mathrm{Hom}(T/\pm 1, \mathbb{G}_m) = \mathbb{Z} \cdot \Phi \subset X^*(T)$$

Note this reconfirms that TSp_{2n} is indeed the adjoint group. On the other hand, all elements of Φ^+ become elements of $X_*(T_{\mathrm{PSp}}) = \mathrm{Hom}(\mathbb{G}_m, T/\pm 1)$. But

$$X_*(T_{\mathrm{PSp}}) = \mathbb{Z}\langle e_1^\vee, e_2^\vee, \cdots e_n^\vee, \frac{e^\vee}{2}\rangle$$

where $\frac{e^\vee}{2} : u \mapsto \mathrm{diag}(\sqrt{u}, \cdots, \sqrt{u}, \sqrt{u}^{-1}, \cdots, \sqrt{u}^{-1})$ and $\sqrt{u}$ is defined up to sign. This is the coweight lattice of $\mathfrak{g}$.

11. (a) $G = \mathrm{SL}_n$. In this case the flag variety consists of all complete flags. Pick B to be the collection of all upper triangular matrices, and T to be the collection of all diagonal matrices. It has Lie algebra $\mathfrak{sl}_n$, with a basis

$$\{E_{ii} - E_{nn} \mid i < n\} \cup \{E_{ij} \mid i \neq j\}$$

Now if $t = (t_1, \cdots, t_{n-1}, t_n = (t_1 \cdots t_{n-1})^{-1}) \in T$ we compute

$$tE_{ij}t^{-1} = \frac{t_i}{t_j}E_{ij}$$

So if we set $e_i : T \to \mathbb{G}_m, t \mapsto t_i$. Then we have root space decomposition

$$\mathfrak{g} = \mathfrak{t} \bigoplus_{i \neq j} \mathfrak{g}_{e_i - e_j}$$

Hence $\Phi = \{e_i - e_j \mid i \neq j\}$, where $e_n = -(e_1 + \cdots + e_{n-1})$. Moreover $X^*(T) = \mathbb{Z}\langle e_1, \cdots, e_{n-1}\rangle$. On the other hand we may set

$$e_i^\vee : \mathbb{G}_m \to T, t \mapsto (1, 1, \cdots, t, \cdots, t^{-1})$$

for $1 \leq i \leq n-1$. We also set $e_n^\vee$ to be the trivial map for simplicity. Then clearly $X_*(T) = \mathbb{Z}\langle e_1^\vee, \cdots, e_{n-1}^\vee\rangle$. To determine the coroots, we note:

$$\varphi_{e_i - e_j} : \mathrm{SL}_2 \to \mathrm{SL}_n, \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \mapsto \mathrm{diag}(1, \cdots, t, \cdots, t^{-1}, \cdots, 1)$$

where t is at the i-th place and t^{-1} is at the j-th place. This tells us

$$(e_i - e_j)^\vee = e_i^\vee - e_j^\vee$$

So in summary

$$X^*(T) = \mathbb{Z}\langle e_1, \cdots, e_{n-1}\rangle; \quad X_*(T) = \mathbb{Z}\langle e_1^\vee, \cdots, e_{n-1}^\vee\rangle$$
$$\Phi = \{e_i - e_j \mid i \neq j\}; \quad \Phi^\vee = \{e_i^\vee - e_j^\vee \mid i \neq j\}$$

where $e_n = -(e_1 + \cdots + e_{n-1})$, $e_n^\vee$ is the trivial map.

The Weyl group is then S_n and the Bruhat decomposition and the description of the line bundle is similar as before. The stablier $P_I = \mathrm{stab}_G(F_I)$ is a blockly upper triangular matrix corresponding to index I. This is a parabolic subgroup of SL_n. Then fibre $\pi_I^{-1}(F_I)$ is the flag variety corresponding to affine algebraic group P_I.

It is also easy to determine the center: $Z_G = \mu_n$ the group of n-th root of unity. This is a finite group, so again $\mathrm{PSL}_n = \mathrm{SL}_n/\mu_n$ is an affine algebraic group. The roots and coroots for PSL_n is the same for SL_n. We only need to compute its characters and cocharacters. In fact

$$X^*(T_{\mathrm{PSL}_n}) = \mathrm{Hom}(T/\mu_n, \mathbb{G}_m) = \mathbb{Z}\Phi$$

and if we set

$$\frac{e^\vee}{n} : \mathbb{G}_m \to T/\mu_n, t \mapsto (t^{\frac{1}{n}}, t^{\frac{1}{n}}, \cdots, t^{-\frac{n-1}{n}})$$

where $t^{\frac{1}{n}}$ is defined up to n-th root of unity. So $n \cdot e^\vee = e_1^\vee + \cdots e_{n-1}^\vee$. We have:

$$X_*(T_{\mathrm{PSL}_n}) = \mathbb{Z}\langle e_1^\vee, e_2^\vee, \cdots, e_{n-1}^\vee, \frac{e^\vee}{n}\rangle$$

(b) $G = \mathrm{SO}_{2n}$. We take our symmetric bilinear form to be $\begin{pmatrix} & I_n \\ I_n & \end{pmatrix}$. Then SO_{2n} is the group of isometries preserving this bilinear form with determinant 1. The flag variety is

$$\mathcal{B} = \{F_1 < F_2 < \cdots < F_{n-1} \mid \dim F_i = i, F_i \leq F_i^\perp\}$$

As in the case of $\mathrm{Sp}(2n)$, we see $\mathcal{B}$ is projective and G acts transitively on $\mathcal{B}$. Now consider

$$\mathcal{B}' = \{F_1 < F_2 < \cdots < F_n \mid \dim F_i = i, F_i \leq F_i^\perp \mid \dim(F_n \cap \langle e_1, \cdots, e_n\rangle) \text{ is even}\}$$

We know the projection map $B' \to \mathcal{B}$ is in fact a G-equivariant isomorphism. Let $F' = \{\langle h_1\rangle < \langle h_1, h_2\rangle < \cdots < \langle h_1, h_2, \cdots, h_n\rangle\}$ and $F = \pi(F')$ be the standard flag. Now set $B = \mathrm{stab}_G(F) = \mathrm{stab}_G(F')$. Note this forces each $b \in B$ to have form $\begin{pmatrix} A & D \\ 0 & C \end{pmatrix}$ where A is upper triangular. But we must have

$$\begin{pmatrix} A^t & \\ D^t & C^t \end{pmatrix} \cdot \begin{pmatrix} & I \\ I & \end{pmatrix} \cdot \begin{pmatrix} A & D \\ & C \end{pmatrix} = \begin{pmatrix} & I \\ I & \end{pmatrix}$$

This forces $C = A^{-t}, D^tC + C^tD = 0$. In particular C is again lower triangular. As in the case of $\mathrm{Sp}(2n)$ we deduce that B is indeed a Borel. We can choose the maximal torus $T = \{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1})\} \simeq \mathbb{G}_m^n$.

The Lie algbera $\mathfrak{g} = \mathfrak{so}(2n)$ consists of matrices $\begin{pmatrix} A & C \\ D & -A^t \end{pmatrix}$ where C, D are skem-symmetric matrices. In particular it has a basis

$$\{E_{i,j} - E_{n+j,n+i}\} \cup \{E_{i,n+j} - E_{j,n+i} \mid i < j\} \cup \{E_{n+i,j} - E_{n+j,i} \mid i < j\}$$

We then compute

$$t(E_{i,j} - E_{n+j,n+i})t^{-1} = \frac{t_i}{t_j}(E_{i,j} - E_{n+j,n+i})$$

$$t(E_{i,n+j} - E_{j,n+i})t^{-1} = t_it_j(E_{i,n+j} - E_{j,n+i})$$

$$t(E_{n+i,j} - E_{n+j,i})t^{-1} = \frac{1}{t_it_j}(E_{n+i,j} - E_{n+j,i})$$

So we have

$$\Phi = \{e_i - e_j \mid i \neq j\} \cup \{e_i + e_j, -e_i - e_j \mid i < j\}$$

where $e_i : T \to \mathbb{G}_m, t \mapsto t_i$ forms a basis for $X^*(T)$, i.e. $X^*(T) = \mathbb{Z}\langle e_1, \cdots, e_n\rangle$. On the other hand, we set $e_i^\vee : \mathbb{G}_m \to T, t \mapsto (1, \cdots, t, \cdots, t^{-1}, \cdots, 1)$. So $X_*(T) = \mathbb{Z}\langle e_1^\vee, \cdots, e_n^\vee\rangle$. As in the case for $\mathrm{Sp}(2n)$, we see:

$$\Phi^\vee = \{e_i^\vee - e_j^\vee \mid i \neq j\} \cup \{e_i^\vee + e_j^\vee, -e_i^\vee - e_j^\vee \mid i < j\}$$

From this we see that $\frac{\pm e_i \pm e_j}{2} \in Y \backslash X^*(T)$, so $\mathrm{SO}(2n)$ is not simply-connected.

Now we need to compute the Weyl group $W = W(\mathrm{SO}(2n))$. As in the case of $\mathrm{Sp}(2n)$ we know:

$$N(T) = \{(a_{ij}) \in \mathrm{SO}_{2n} \text{ monoidal} \mid a_{ij} \neq 0 \iff a_{i+n,j+n} \neq 0\}$$

Now if $g = \sum a_j E_{\sigma(j),j} \in N(T)$ for $a_j \neq 0, \sigma(j+n) = \sigma(j) + n$. We replace g by gt for some $t \in T$, and we may assume $a_j = 1$ for $1 \leq j \leq n$. We now impose the condition

$$g^t \begin{pmatrix} & I_n \\ I_n & \end{pmatrix} g = \begin{pmatrix} & I_n \\ I_n & \end{pmatrix}$$

We then see: $a_{j+n}a_j = 1$ for each j. In particular, it forces $a_j = 1$ for all j. Moreover we need to impose the determinant 1 condition, so

$$W = N(T)/T \simeq \{\sigma \in A_{2n} \mid \sigma(i) + n = \sigma(i+n)\}$$

The proof for Bruhat decomposition is done before. The interpretation for line bundles is uniform.

Now we wish to describe the parabolic subgroups explicitly. We take the model $\mathcal{B}'$ for the flag variety. Notice that there exists another

$$\mathcal{B}'' = \{F_1 < F_2 < \cdots < F_n \mid \dim F_i = i, F_i \leq F_i^\perp \mid \dim(F_n \cap \langle e_1, \cdots, e_n\rangle) \text{ is odd}\}$$

For each $I \subset \{1, 2, \cdots, n\}$, there exists $\pi'_I : \mathcal{B}' \to \mathcal{P}'_I$ and $\pi''_I : \mathcal{B}'' \to \mathcal{P}''_I$. The stablizers $P'_I = \mathrm{stab}_G(F'_I)$ and $P''_I = \mathrm{stab}_G(F''_I)$ are subgroups containing B. Note runs through the following distinct subgroups:

$$\{P'_I, P''_I \mid n-1 \notin I\}$$

There are in total $2 \times 2^{n-1} = 2^n$ of them, and hence they form a complete collection of parabolic subgroups.

Now let's study the center. Again we know Z_G consists of only diagonal matrices. By testing this with Weyl group elements, we see $Z_G = \{\pm I\}$. Hence we have the adjoint group $\mathrm{PSO}_{2n} = \mathrm{SO}_{2n}/\{\pm I\}$. We need to calculate the characters and cocharacters. But again, $X^*(T_{\mathrm{PSO}}) = \mathbb{Z}\Phi^\vee$, and $X_*(T_{\mathrm{PSO}}) = \mathbb{Z}\langle e_1^\vee, \cdots, e_n^\vee, \frac{e_1^\vee + e_2^\vee + \cdots + e_n^\vee}{2}\rangle$.

(c) $G = \mathrm{SO}_{2n+1}$. We take our symmetric bilinear form to be $\begin{pmatrix} & I_n & \\ I_n & & \\ & & 1 \end{pmatrix}$, with respect to basis $\{h_1, \cdots, h_n, f_1, \cdots, f_n, r\}$. The flag variety is

$$\mathcal{B} = \{F_1 < F_2 < \cdots < F_n \mid \dim F_i = i, F_n \subset F_n^\perp\}$$

As before $\mathcal{B}$ is projective and G acts transitively on it. Let

$$F = \{\langle h_1\rangle < \langle h_1, h_2\rangle < \cdots < \langle h_1, h_2, \cdots, h_n\rangle\}$$

be the standard flag. Then $\mathrm{stab}_G(F)$ consists of matrices of form $\begin{pmatrix} A & C & E \\ 0 & D & H \\ 0 & J & K \end{pmatrix}$ for A upper triangular invertible, and it must preserves the symmetric bilinear form. Some computations show:

$$D = A^{-t}, H = 0.$$

So we can rearrange the basis to make $\mathrm{stab}_G(F)$ upper triangular. So $B = \mathrm{stab}_G(F)$ is a Borel as before, and the diagonal elements are torus:

$$T = \{\mathrm{diag}(t_1, \cdots, t_n, t_1^{-1}, \cdots, t_n^{-1}, 1)\}$$

We have canonical choices for a basis of characters and cocharacters:

$$X^*(T) = \mathbb{Z}\langle e_1, \cdots, e_n\rangle; \quad X_*(T) = \mathbb{Z}\langle e_1^\vee, \cdots, e_n^\vee\rangle$$

We need to look at the Lie algebra $\mathfrak{so}_{2n+1}$, it consits of matrices:

$$\begin{pmatrix} A & C & H \\ D & -A^t & I \\ -I^t & -H^t & 0 \end{pmatrix}$$

for C, D skew-symmetric. So it has a basis:

$$\{E_{ij} - E_{n+j,n+i}\} \cup \{E_{i,n+j} + E_{j,n+i} \mid i < j\} \cup \{E_{i+n,j} + E_{j+n,i} \mid i < j\}$$
$$\cup \{E_{i,2n+1} - E_{2n+1,n+i}, \mid 1 \leq i \leq n\} \cup \{E_{i+n,2n+1} - E_{2n+1,i} \mid 1 \leq i \leq n\}$$

It corresponds to roots:

$$\Phi = \{e_i - e_j \mid i \neq j\} \cup \{e_i + e_j, -e_i - e_j \mid i < j\} \cup \{\pm e_i \mid 1 \leq i \leq n\}$$

and coroots

$$\Phi^\vee = \{e_i^\vee - e_j^\vee \mid i \neq j\} \cup \{e_i^\vee + e_j^\vee, -e_i^\vee - e_j^\vee \mid i < j\} \cup \{\pm 2e_i^\vee \mid 1 \leq i \leq n\}$$

Note $\mathbb{Z}\Phi = X^*(T)$, this shows $\mathrm{SO}(2n+1)$ is the adjoint group, hence it has trivial center.

To compute the Weyl group, we note

$$N(T) = \{g = (a_{ij}) \in \mathrm{SO}_{2n+1} \text{ monoidal} \mid a_{2n+1,2n+1} \neq 0\}$$

This gives us identification

$$N(T)/T \simeq \{\sigma \in S_{2n} \mid \sigma(n+i) = \sigma(n) + i\}$$

(One can also see this by noting that this root system is dual to that of $\mathrm{Sp}(2n)$). The proof of Bruhat decomposition and the construction of parabolic subgroups is standard.

□